ShadowGDNet: Gated and Decoupled Fusion of Unreliable Scene Priors for Mask-Free Shadow Removal

Supplementary Material

This document gives the material referenced in the main paper: the architecture details, the prior maps, the prior reliability protocol, further ablations, the depth model study on all datasets, additional analyses, the cost breakdown and the challenge leaderboard. Unless stated otherwise, all results are on the 100-image WSRD+ [11] validation set.

1. Architecture Specification

Table 1 lists the attention type and head split of each stage (Sec. 4.5 of the main paper). Prior modulation is only used in the three coarsest stages, the same blocks in which DenseSR uses its joint modulation; the other stages keep DenseSR's channel attention blocks. These have $N_h \geq 4$, so the split is always $(1/4, 1/4, 1/2)$, and the rules for $N_h \leq 2$ are never used in this backbone.

Keeping the layout of DenseSR means that the comparison changes only how the priors are combined. The layout also suits the method for three further reasons. Three of the four channel-attention stages are configured with only one or two heads, too few for three head groups. The semantic features have a patch stride of 16, so at full resolution they add little to the RGB features, and the same holds for smooth depth. Finally, every window needs its own $N \times N$ prior maps, and the full-resolution stage has 64 times as many windows as the bottleneck. Other placements of DMMA are left for future work.

Stage	**Scale**	N_h	**Attention**	**Split**
Encoder 1	H	1	Channel	—
Encoder 2	$H/2$	2	Channel	—
Encoder 3	$H/4$	4	Prior-modulated	1/1/2
Bottleneck	$H/8$	16	Prior-modulated	4/4/8
Decoder 1	$H/4$	8	Prior-modulated	2/2/4
Decoder 2	$H/2$	4	Channel	—
Decoder 3	H	2	Channel	—

Table 1. Where prior-modulated attention is used. Split gives the number of unmodulated, semantic and geometric heads. Channel-attention stages are listed with the head count of their configuration, which their blocks do not use.

2. Prior Maps

This section defines the two maps used in Sec. 4.5 of the main paper. Both are computed inside a window of $N = \text{win_size}^2$ tokens.

The semantic map $M_{\text{sem}} \in \mathbb{R}^{B \cdot n_W \times 1 \times N \times N}$ uses the DINOv3 feature $\mathbf{s}_i \in \mathbb{R}^d$ of each token i in the window. We L2-normalize the features, compute pairwise cosine similarities and clamp them at zero:

$$\hat{\mathbf{s}}_i = \frac{\mathbf{s}_i}{\|\mathbf{s}_i\|_2 + \epsilon}, \quad M^{ij}_{\text{sem}} = \max\left(\hat{\mathbf{s}}_i \cdot \hat{\mathbf{s}}_j^T,\, 0\right). \tag{1}$$

Tokens on the same object or material get high values, and tokens on different surfaces get low values.

The geometric map $M_{\text{geo}} \in \mathbb{R}^{B \cdot n_W \times 1 \times N \times N}$ uses the 3D point $\mathbf{p}_i \in \mathbb{R}^3$ and surface normal $\mathbf{n}_i \in \mathbb{R}^3$ of each token. The distance of token j from the plane through token i is the absolute dot product of their position difference with $\mathbf{n}_i$, and we average it over both directions:

$$d_{ij} = \left|(\mathbf{p}_i - \mathbf{p}_j) \cdot \mathbf{n}_i\right|, \quad \tilde{d}_{ij} = \frac{d_{ij} + d_{ji}}{2}, \tag{2}$$

$$M^{ij}_{\text{geo}} = \exp\left(-\tilde{d}_{ij}\right). \tag{3}$$

Tokens on the same surface get high values, tokens at different depths get lower values, and the averaging makes the map symmetric.

The exponent has no units. DepthPro predicts metric depth, and we divide each depth map by its median over valid pixels before back-projection. The points $\mathbf{p}_i$ are then in units of the median scene depth, so M_{geo} does not depend on the absolute scale of the scene. The scale error in our robustness tests is applied after this normalization (Sec. 12), so it does change M_{geo}. M_{geo} also changes with the assumed field of view, which sets the ratio between lateral and depth extent; Sec. 7 measures the effect on the results. The normals $\mathbf{n}_i$ are computed from the point map as $\mathbf{n} \propto \partial_x \mathbf{P} \times \partial_y \mathbf{P}$ with finite differences and normalized to unit length.

Both maps scale the attention logits before the softmax. For a head group g with map M_g and the relative position bias B of [9],

$$\mathbf{A}_g = \text{softmax}\left(\left(\tfrac{1}{\sqrt{d_h}}\mathbf{Q}_g\mathbf{K}_g^\top\right) \odot M_g \;+\; B\right), \tag{4}$$

where $\odot$ is element-wise multiplication, $M_g = M_{\text{sem}}$ for the semantic group, M_{geo} for the geometric group, and $M_g \equiv 1$ for the unmodulated group. The baseline applies $M_{\text{sem}} \odot M_{\text{geo}}$ to every head. Since the maps scale the logits and not the attention weights, a map near zero removes the content term, and the attention of that group becomes $\text{softmax}(B)$, which depends only on the relative

positions of the tokens. The outputs of the groups are concatenated and passed through the shared output projection. This projection is linear, so the attention output is the sum of the projected outputs of the three groups, and a corrupted map changes only the term of its own group. This holds within the attention block; depth also enters through the input channel and semantic features at the bottleneck.

3. Prior Reliability Inside and Outside Shadows

Table 2 gives the measurements behind Table 1 of the main paper. We compare two predictions, not a prediction with ground truth. ISTD+ has no dense depth or normals, so the prior extracted from the shadow-free image serves as the reference, and we measure how far the prior from the shadowed image is from it. The numbers are consistency measures, not accuracies. Since both images show the same scene, the true depth and normals are the same in both, so the discrepancy is a lower bound on the sum of the two prediction errors.

ISTD+ shadow and shadow-free images are taken from a fixed viewpoint and are pixel-aligned, so we apply no registration. DepthPro estimates the metric scale separately for each image, so the two predictions can differ by a global factor. We first align the shadowed-image depth to the shadow-free depth with a single factor, the ratio of their medians over valid pixels. Without this step the difference would mostly reflect the global scale. Normals are computed from each depth map in the same way as in the network, with the same field of view. Pixels where either depth map is invalid or non-positive are excluded. We erode the shadow mask by 5 px so that boundary pixels, where alignment errors are largest, are not counted in either region. Depth discrepancy is the median absolute relative difference over the region, normal discrepancy is the mean angle between corresponding normals, and semantic consistency is the mean cosine similarity between DINOv3 features at the same positions; we use the median for depth and the mean for the other two measures. The main paper also reports that the depth discrepancy correlates with shadow intensity ($r = 0.46$) and is higher near soft boundaries than near hard ones (0.169 vs. 0.118). Two confounds are not controlled: shadows in ISTD+ often fall on weakly textured ground, where depth may be less reliable even without a shadow, and the ISTD+ shadow-free images are color-corrected. We have no matched-texture control.

4. Component Ablation with SSIM

Table 3 repeats the component ablation of the main paper with SSIM. The baseline is DenseSR retrained under our protocol. Each module improves PSNR on its own, and every gap between the baseline and the full model is several times the seed standard deviation. The gain varies by more than a factor of five across datasets, from +0.77 dB on ISTD to +0.14 dB on WSRD+. On ISTD, SSIM decreases slightly for GSF-only, DMMA-only and the full model (by 0.001 to 0.002), within the seed standard deviation. On ISTD+ and WSRD+ the SSIM gains are large relative to the PSNR gains; we do not interpret this difference further.

Discrepancy	**Non-shadow**	**Shadow**	**Ratio**
Depth, relative ↓	0.021	0.144	6.9×
Normals, angular (°) ↓	4.8	19.6	4.1×
Semantic cosine sim. ↑	0.972	0.883	0.91×

Table 2. Consistency of each prior between the shadowed and the shadow-free ISTD+ [7] image, split by an eroded shadow mask.

5. Design Ablations

The two studies in this section were used to choose the configuration. Both were run on the WSRD+ validation set, which is also the set we report, so the WSRD+ numbers of the chosen configuration may be slightly optimistic. The chosen configuration was then used unchanged on ISTD and ISTD+, whose test sets were not used for these choices.

5.1. GSF at Different Scales

To see where the skip features hurt most, we enable GSF at each skip separately, with DMMA turned off (Table 4). These are single runs. Gating the $H/2$ skip alone gives the largest gain (26.33 dB). Both pairs we tested reach 26.34 dB, and gating all three skips gives the best GSF-only result, 26.35 dB, so the third gate adds little. We gate all three skips, which gave the best single-run result, although all differences in this table (0.01 to 0.07 dB) are within the seed variation of the main paper.

5.2. Head Split

Table 5 varies how the heads are split between the unmodulated, semantic and geometric groups, with and without GSF. Where a proportion does not divide a layer's head count, the group sizes are rounded per layer. These are single runs, so differences between neighboring rows should be compared with the seed standard deviation of 0.02 to 0.06 dB in the main paper. Most decoupled splits beat joint modulation, and the geo-heavy split, which assigns half of the heads to depth and normals, was best in this run. With GSF enabled, the RGB-heavy split ties with joint modulation and the no-semantic split falls 0.01 dB below it. All differences between decoupled splits are within seed variation, so we do not draw conclusions about the role of individual groups. The order of the splits is similar with and without GSF. The GSF-enabled joint and geo-heavy rows correspond to the GSF-only and full models, but they are

Configuration	GSF	DMMA	ISTD [12]		ISTD+ [7]		WSRD+ (val) [11]	
			PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Baseline [8]	×	×	30.64±.05	.976±.002	33.98±.06	.974±.003	26.28±.03	.838±.004
+ GSF only	✓	×	30.97±.04	.975±.002	34.17±.05	.978±.002	26.34±.02	.851±.003
+ DMMA only	×	✓	31.08±.05	.975±.002	34.24±.06	.981±.002	26.36±.03	.855±.003
Ours (both)	✓	✓	**31.41±.04**	.974±.002	**34.43±.05**	**.985±.002**	**26.42±.02**	**.865±.003**

Table 3. Component ablation with PSNR and SSIM, mean ± standard deviation over three seeds.

Config.	GSF_H	$\text{GSF}_{H/2}$	$\text{GSF}_{H/4}$	PSNR↑	SSIM↑
Baseline [8]	×	×	×	26.28	0.838
H only	✓	×	×	26.29	0.840
$H/2$ only	×	✓	×	26.33	0.847
$H/4$ only	×	×	✓	26.29	0.840
H+$H/2$	✓	✓	×	26.34	0.849
$H/2$+$H/4$	×	✓	✓	26.34	0.850
All scales	✓	✓	✓	**26.35**	**0.851**

Table 4. GSF at different skip scales on WSRD+ [11] validation, single runs, with DMMA disabled.

Head Split	GSF disabled		GSF enabled	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Baseline (joint) [8]	26.28	0.838	26.35	0.851
Equal split	26.33	0.850	26.39	0.861
RGB-heavy	26.29	0.843	26.35	0.855
Sem-heavy	26.31	0.847	26.37	0.858
No RGB heads	26.30	0.845	26.36	0.856
No Sem heads	26.28	0.842	26.34	0.853
Geo-heavy	**26.37**	**0.855**	**26.42**	**0.865**

Table 5. Head split on WSRD+ [11] validation, single runs, with and without GSF. Group sizes are rounded per layer where needed; geo-heavy $(1/4, 1/4, 1/2)$ is the split of Sec. 4.5 of the main paper and the one we use.

single runs and not three-seed means, which is why they read 26.35 and 26.42 here and 26.34 and 26.42 in the main paper. Because the geo-heavy split was selected as the best single run on this set, its single-run score is subject to selection bias.

Table 6 shows joint modulation and four decoupled splits with GSF enabled. Three of them appear in Table 5 (equal, RGB-heavy and geo-heavy); the $(1/6, 1/3, 1/2)$ split is an additional single run. Three of the four beat joint modulation, and the RGB-heavy split ties with it at 26.35 dB. The geo-heavy split is best, 0.07 dB above joint modulation. As these are single runs, differences between neighboring splits are within the seed variation.

RGB	Sem	Geo	PSNR↑	SSIM↑
—	—	—	26.35	0.851
1/3	1/3	1/3	26.39	0.861
1/2	1/4	1/4	26.35	0.855
1/6	1/3	1/2	26.40	0.862
1/4	**1/4**	**1/2**	**26.42**	**0.865**

Table 6. Head split with GSF enabled on WSRD+ validation, single runs. The first row is joint modulation.

6. Depth Model on All Datasets

The main paper reports the swap of architecture and depth model on WSRD+. Table 7 gives the same study on all three datasets. DenseSR uses the metric-depth model of DAv2, which is median-normalized and back-projected exactly like DepthPro. The WSRD+ pattern holds: with the same depth model, our architecture beats the baseline by a margin several times larger than the gain from giving the baseline a better depth model.

6.1. Factorial Breakdown

The swap is a 2×2 factorial design over architecture and depth model, so the two effects can be separated. With WSRD+ PSNRs of 26.28 and 26.31 for the baseline with DAv2 and DepthPro, and 26.38 and 26.42 for ours, the main effect of the architecture is

$$\tfrac{1}{2}(26.38 + 26.42) - \tfrac{1}{2}(26.28 + 26.31) = +0.105 \text{ dB}, \quad (5)$$

the main effect of the depth model is

$$\tfrac{1}{2}(26.31 + 26.42) - \tfrac{1}{2}(26.28 + 26.38) = +0.035 \text{ dB}, \quad (6)$$

and the interaction is $(26.42 - 26.38) - (26.31 - 26.28) = +0.010$ dB. Each cell is a single run and the depth effect is below the seed variation, so the ratio between the two effects is not reliable; across the three datasets the architecture effect ranges from 0.10 to 0.72 dB and the depth effect from 0.03 to 0.07 dB (Table 7). We therefore do not run an F-test. With three seeds per cell, a two-way analysis of variance would give the same breakdown with uncertainty attached, and this is the natural next step for this experiment.

Architecture	Depth prior	ISTD [12]		ISTD+ [7]		WSRD+ (val) [11]	
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
DenseSR [8]	DAv2 [15]	30.64	0.976	33.98	0.974	26.28	0.838
DenseSR [8]	DepthPro [2]	30.69	0.977	34.04	0.976	26.31	0.841
Ours	DAv2 [15]	31.35	0.973	34.36	0.984	26.38	0.859
Ours	**DepthPro** [2]	**31.41**	0.974	**34.43**	**0.985**	**26.42**	**0.865**

Table 7. Swap of architecture and depth model on all three datasets.

Assumed FOV	45°	60°	75°
DenseSR [8]	26.26	26.28	26.25
Ours	26.39	**26.42**	26.38
Margin	0.13	0.14	0.13

Table 8. PSNR on WSRD+ validation for different assumed fields of view.

	Mean Δ	**95% CI**	p	**Improved**
ISTD	0.77	[0.63, 0.91]	<1e-6	478/540
ISTD+	0.45	[0.36, 0.55]	<1e-6	441/540
WSRD+	0.14	[0.09, 0.19]	3.1e-4	71/100

Table 9. Per-image PSNR difference (dB) between ShadowGDNet and DenseSR, with bootstrap intervals and Wilcoxon signed-rank p-values.

7. Field-of-View Sensitivity

The geometric prior depends on the assumed field of view, so we also vary it. Table 8 repeats the WSRD+ comparison at 45°, 60° and 75°. Both models change by at most 0.04 dB and the margin stays at 0.13 to 0.14 dB, so the result does not depend on the 60° choice, although the geometry itself remains approximate.

8. Paired Significance Analysis

Table 9 gives the per-image analysis summarized in the main paper. On WSRD+ the interval excludes zero and 71 of the 100 images improve, although the ten most improved images account for 41% of the total gain.

9. Performance by Shadow Size

Table 10 splits the WSRD+ validation set into quartiles by shadow area, with 25 images each. The gain increases with shadow area, from +0.07 to +0.23 dB. This is the direction the motivation predicts: a larger shadow means more affected encoder features and a larger area where the priors are unreliable. With 25 images per quartile we report means only, without intervals. Per-image gains are heavy-tailed (the ten most improved images account for 41% of the total gain), so these quartile means may be driven by a few images.

Shadow area	**DenseSR**	**Ours**	**Δ**
Q1 (smallest)	27.91	27.98	+0.07
Q2	26.84	26.95	+0.11
Q3	25.77	25.94	+0.17
Q4 (largest)	24.62	24.85	+0.23

Table 10. PSNR by shadow-area quartile on WSRD+ validation, 25 images per quartile.

Scale	**In shadow**	**Outside**	**Ratio**
H	0.34	0.71	0.48
$H/2$	0.29	0.68	0.43
$H/4$	0.26	0.66	0.39

Table 11. Mean GSF gate value inside and outside the shadow mask at each skip scale, on ISTD+ [7].

Splitting the same set by boundary type gives +0.11 dB for scenes with mostly hard boundaries and +0.18 dB for scenes with mostly soft penumbrae, consistent with the priors being least reliable where the boundary is diffuse. These are two means without intervals, and the split uses a hand-set threshold on penumbra width.

10. Gate Statistics

Table 11 gives the mean gate value inside and outside the ground-truth shadow mask at each skip scale, computed on ISTD+ in evaluation mode, so batch normalization uses its running statistics. The gates are lower inside shadows at every scale, as the motivation in the main paper assumes. The gate thus responds to the shadow without ever being shown a mask, although, since shadow regions are also darker, it may partly respond to brightness. Testing whether it causes the accuracy gain, for example by replacing it with a constant, a spatially shuffled or an inverted gate at inference, is a natural next step.

11. Full Noise Sweep

Table 12 gives the full Gaussian noise sweep summarized in the main paper, including the intermediate noise levels. As in the main paper, we compare the complete models, DenseSR with DAv2 depth and ShadowGDNet with Depth-

σ	DenseSR [8]		ShadowGDNet (ours)	
	PSNR↑	Δ	PSNR↑	Δ
0.00	26.28	—	26.42	—
0.05	26.04	−0.24	26.34	−0.08
0.10	25.74	−0.54	26.24	−0.18
0.15	25.42	−0.86	26.13	−0.29
0.20	25.07	−1.21	26.00	−0.42
0.30	24.33	−1.95	25.71	−0.71

Table 12. PSNR under Gaussian noise on the geometric priors at test time, on WSRD+ [11] validation.

Corruption	DenseSR	Ours	Ratio
i.i.d. Gaussian, σ=0.2	−1.21	−0.42	2.9×
Spatially correlated noise	−1.44	−0.51	2.8×
Boundary displacement, 5 px	−1.02	−0.37	2.8×
Missing depth, 10% blocks	−1.63	−0.55	3.0×
Depth blur	−0.74	−0.28	2.6×
Scale error, ±20%	−0.58	−0.31	1.9×
Noise inside shadow only	−1.38	−0.44	3.1×

Table 13. PSNR change on WSRD+ validation under structured corruption of the geometric priors at test time.

Pro depth, so the sweep measures the robustness of ShadowGDNet as a whole. The noise is added after median normalization to the point map and the normal map. The depth input channel is the z-coordinate of the point map, so it receives the same noise, and normals are perturbed directly rather than recomputed. Since σ is relative to each model's normalized depth (DAv2 for DenseSR, DepthPro for ours), the noise is matched in normalized units, not in metric error.

12. Structured Prior Corruption

Table 13 gives all structured corruptions, including depth blur. The strength of each corruption is set so that its mean error on the normalized depth equals that of the σ=0.2 Gaussian case. All corruptions are applied after median normalization, in the same way as the Gaussian noise. The advantage holds for every corruption and is largest for missing depth and for noise inside the shadow. The exception is the scale error. Because it is applied after normalization, it multiplies every $\tilde{d}_{ij}$ in M_{geo} by the same factor, which makes the map sharper or flatter but keeps the order of its scores, and the ratio falls to 1.9×. Apart from this case, the ratio is similar across very different corruption types.

13. Corrupting Each Prior Separately

We also corrupt the semantic prior alone, the geometric prior alone, and both, at $\sigma = 0.2$.

Table 14 shows the result. DenseSR loses a similar amount whichever prior is corrupted (0.98 and 1.21 dB).

Corrupted prior	Heads hit	DenseSR [8]	Ours
None	—	26.28	26.42
Semantic only	1/4	−0.98	−0.35
Geometric only	1/2	−1.21	−0.42
Both	3/4	−1.86	−0.71

Table 14. Corrupting each prior separately at σ=0.2 on WSRD+ [11] validation, as PSNR change from the clean row. Heads hit is the fraction of attention heads whose modulation is affected under DMMA.

Method	Params	MACs @256^2
BMNet [16] (mask-based)	*0.4M*	*11G*
DCShadowNet [6]	*10.6M*	*38G*
ShadowFormer [5] (mask-based)	*11.4M*	*52G*
DHAN [3] (mask-based)	*21.8M*	*105G*
DenseSR [8]	*24.74M*	*92.0G*
ShadowGDNet (ours)	**24.78M**	**92.4G**
OmniSR [13]	*35.1M*	*121G*
ShadowRefiner [4]	*61.9M*	*178G*
StableShadowDiffusion [14]	*1.07B*	*6.8T*

Table 15. Cost of the trainable network for the compared methods. Numbers in *italics* are published or derived, not measured by us.

ShadowGDNet loses about a third as much in every case (0.35, 0.42 and 0.71 dB); its loss grows with the number of affected head groups, but not in proportion to it. Neither model is additive when both priors are corrupted. These are the complete models, which also differ in GSF and in the depth model, and the priors also reach the network through the input channel and the bottleneck. Like the other corruption results, the table therefore describes the full model; separating the roles of DMMA and GSF under corruption is left for future work.

14. Computational Cost

The trainable network and the full inference pipeline differ in cost by more than an order of magnitude, so we report them separately. Table 15 compares our trainable network with the methods in the main paper. Parameter count does not predict accuracy here: the two smallest models and the largest one all score below mid-sized models, and our count is close to DenseSR because the two modules add only 0.04M parameters. Counts are multiply-accumulates for one 256^2 pass, except for StableShadowDiffusion, which is reported over 20 sampling steps. Frozen prior extractors are not included in this table.

Table 16 splits our pipeline into trainable and frozen parts. The two modules add 0.04M parameters and 0.4G multiply-accumulates, which is 0.16% and 0.43% of the

Component	Params	MACs @256^2
DenseSR backbone [8] (trainable)	24.74M	92.0G
+ GSF, three gates	+0.04M	+0.4G
+ DMMA	+0.00M	+0.00G
ShadowGDNet (trainable)	**24.78M**	**92.4G**
Frozen DINOv3-L/16 [10]	0.30B	82G
Frozen DepthPro [2]	0.95B	7.05T
Full pipeline	1.27B	7.2T

Table 16. Cost of our pipeline split into trainable and frozen parts, in multiply-accumulates per image. The trainable network and DINOv3 are counted at 256^2, DepthPro at its fixed internal resolution of 1536^2.

trainable backbone, and DMMA adds no parameters because it only splits existing heads. The frozen extractors account for 1.25B of the 1.27B pipeline parameters and about 7.1T of its 7.2T multiply-accumulates. Almost all of the pipeline cost lies in the frozen extractors.

Two details matter for reproduction. We count multiply-accumulates, not twice that number; profiling tools differ on this convention. The frozen model sizes are the published numbers for DINOv3-L/16 and DepthPro. We compute their cost analytically, counting the attention and MLP terms of each 24-layer, 1024-dimensional ViT-L but not layer normalization, patch embeddings or decoders, so these are lower bounds. For DINOv3 we use 261 tokens at 256^2 (256 patches at stride 16, plus one class token and four register tokens). For DepthPro we use the official configuration, which resizes the image to 1536^2. Its patch encoder runs on 35 overlapping 384^2 crops from a three-level image pyramid, and its image encoder and field-of-view encoder each run once on a 384^2 version of the image. Each of these 37 passes has 577 tokens and costs about 0.19T MACs, so DepthPro needs about 7.05T MACs per image.

Because the frozen extractors are about fifty times larger than the trainable network, every efficiency statement in the paper refers to the trainable network only. The depth swap in Sec. 6 shows that our gain remains with the lighter DAv2 prior, so the architectural benefit is available at the same pipeline cost as DenseSR. The priors are computed once per image and can be cached.

14.1. Runtime and Memory

Table 17 reports the runtime and memory of the trainable network: the median latency over 100 forward passes after 20 warm-up passes, and the peak allocated (not reserved) CUDA memory. We did not profile the frozen extractors. We run them offline once per image and cache their outputs, for training and for evaluation, so the trainable network is the only part that runs in each forward pass.

Component	Params	MACs	ms	GB
Trainable network	24.78M	92.4G	18	1.4
Frozen DINOv3-L/16	0.30B	82G	—	—
Frozen DepthPro	0.95B	7.05T	—	—

Table 17. Inference cost at 256^2, batch size one, single A40, fp32. The frozen extractors run offline and were not profiled (—); the DepthPro MACs are per image at its 1536^2 internal resolution.

Team	PSNR↑	SSIM↑	LPIPS↓	Final Rank↓
AIT_TUM	26.680	0.8740	0.0578	1
RAS	26.137	0.8657	0.0708	2
SNU-ISPL-B	25.943	0.8665	0.0849	3
APRIL-AIGC	26.449	0.8481	0.0786	4
DiogenesCask	26.432	0.8677	0.0856	5
SNUCV	25.300	0.8638	0.0676	6
(ours)	**25.639**	**0.8636**	**0.0850**	**7**

Table 18. Final ranking of the NTIRE 2026 Shadow Removal Challenge, scored by the challenge server on the hidden test set. These scores are not comparable to the validation results in the main paper.

15. Challenge Leaderboard

Table 18 gives the final ranking of the NTIRE 2026 Shadow Removal Challenge on the hidden test set. These scores cannot be compared with the validation results in the main paper. Our entry placed seventh with 25.639 dB, 1.04 dB below the winner. It is a single 24.78M-parameter model trained only on the 886 challenge images, with no ensembling, no test-time augmentation and no extra shadow-removal training data. It does use the pretrained frozen extractors DINOv3-L/16 and DepthPro.

The winning entry uses a three-stage cascade with cosine-annealed checkpoint ensembling [1]; we did not verify the pipelines of the other entries. Our submission is a controlled architectural study, while the winning entry is tuned for the leaderboard, so the ranking answers a different question. The comparison that matters for our claim is against the same backbone: two modules improve a prior-guided backbone at almost the same trainable size, and the improvement is larger when the priors are corrupted.

References

[1] Lorenzo Beltrame, Jules Salzinger, Filip Svoboda, Jasmin Lampert, Phillipp Fanta-Jende, Radu Timofte, and Marco Körner. Winner of CVPR2026 NTIRE challenge on image shadow removal: Semantic and geometric guidance for shadow removal via cascaded refinement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026. 6

[2] Alexey Bochkovskiy, Amaël Delaunoy, Hugo Germain, Marcel Santos, Yichao Zhou, Stephan R. Richter, and

Vladlen Koltun. Depth Pro: Sharp monocular metric depth in less than a second. In *International Conference on Learning Representations (ICLR)*, 2025. 4, 6

[3] Xiaodong Cun, Chi-Man Pun, and Cheng Shi. Towards ghost-free shadow removal via dual hierarchical aggregation network and shadow matting GAN. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 10680–10687, 2020. 5

[4] Wei Dong, Han Zhou, Yuqiong Tian, Jingke Sun, Xiaohong Liu, Guangtao Zhai, and Jun Chen. ShadowRefiner: Towards mask-free shadow removal via fast Fourier transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 6208–6217, 2024. 5

[5] Lanqing Guo, Siyu Huang, Ding Liu, Hao Cheng, and Bihan Wen. ShadowFormer: Global context helps shadow removal. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 710–718, 2023. 5

[6] Yeying Jin, Aashish Sharma, and Robby T. Tan. DC-ShadowNet: Single-image hard and soft shadow removal using unsupervised domain-classifier guided network. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5027–5036, 2021. 5

[7] Hieu Le and Dimitris Samaras. Shadow removal via shadow image decomposition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8578–8587, 2019. 2, 3, 4

[8] Yu-Fan Lin, Chia-Ming Lee, and Chih-Chung Hsu. DenseSR: Image shadow removal as dense prediction. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM)*, pages 7026–7035, 2025. 3, 4, 5, 6

[9] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin Transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021. 1

[10] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seung Eun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3. *Transactions on Machine Learning Research*, 2026. 6

[11] Florin-Alexandru Vasluianu, Tim Seizinger, and Radu Timofte. WSRD: A novel benchmark for high resolution image shadow removal. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 1826–1835, 2023. 1, 3, 4, 5

[12] Jifeng Wang, Xiang Li, and Jian Yang. Stacked conditional generative adversarial networks for jointly learning shadow detection and shadow removal. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1788–1797, 2018. 3, 4

[13] Jiamin Xu, Zelong Li, Yuxin Zheng, Chenyu Huang, Renshu Gu, Weiwei Xu, and Gang Xu. OmniSR: Shadow removal under direct and indirect lighting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 8887–8895, 2025. 5

[14] Jiamin Xu, Yuxin Zheng, Zelong Li, Chi Wang, Renshu Gu, Weiwei Xu, and Gang Xu. Detail-preserving latent diffusion for stable shadow removal. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7592–7602, 2025. 5

[15] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 21875–21911, 2024. 4

[16] Yurui Zhu, Jie Huang, Xueyang Fu, Feng Zhao, Qibin Sun, and Zheng-Jun Zha. Bijective mapping network for shadow removal. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5627–5636, 2022. 5